

Epidemiology, Virology, and Pathogenesis of the Zika Virus: From Neglected Tropical Disease to a Focal Point of International Attention.

PMID: 27618294 | Indexed in PubMed

Over the past year, the Zika virus, an arthropod-borne Flavivirus, has transitioned from a relatively unknown tropical disease to the cause of a public health emergency. The Zika virus is transmitted by the *Aedes* species of mosquito as well as by sexual intercourse. Although the symptoms of acute Zika virus infection are usually mild and self-limited, it causes fetal microcephaly in pregnant women, and is associated with an increased risk of Guillain-Barré syndrome. The risk of microcephaly from Zika virus infection is estimated to be highest in women who are infected during the first trimester of pregnancy. The Zika virus has been shown to have significant neurotrophism in vivo and in vitro, although further study is needed to characterize its mechanisms of pathogenesis. Zika virus has previously caused two known outbreaks in the Pacific region prior to the current epidemic in South and Central America, and the current epidemic has affected at least 440,000 to 1,300,000 people. The population of the vector for the current epidemic, *Aedes aegypti*, varies seasonally in the United States, however there have been few documented cases of local spread of the Zika infection in the United States and it is unclear whether epidemic spread of Zika will occur within the United States.

Vector competence of *Aedes aegypti* and *Culex quinquefasciatus* from the metropolitan area of Guadalajara, Jalisco, Mexico for Zika virus.

PMID: 31740702 | Indexed in PubMed

Zika virus (ZIKV) is a mosquito-borne pathogen discovered in the late 40's in Uganda during a surveillance program for yellow fever. By 2014 the virus reached Eastern Island in the Americas, and two years later, the virus spread to almost all countries and territories of the Americas. The mosquito *Aedes aegypti* has been identified as the main vector of the disease, and several researchers have also studied the vector competence of *Culex quinquefasciatus* in virus transmission. The aim of the present study was to evaluate the vector competence of *Ae. aegypti* and *Cx. quinquefasciatus* in order to understand their roles in the transmission of ZIKV in Guadalajara, Jalisco, Mexico. In blood feeding laboratory experiments, we found that *Ae. aegypti* mosquitoes showed to be a competent vector able to transmit ZIKV in this area. On the other hand, we found that F0 *Cx. quinquefasciatus* mosquitoes are refractory to ZIKV infection, dissemination and transmission.

New Paradigms for Virus Detection, Surveillance and Control of Zika Virus Vectors in the Settings of Southeast Asia.

PMID: 27679623 | Indexed in PubMed

Zika virus (ZIKV) has now become a global public health concern. The vectors for ZIKV are *Aedes aegypti* and *A. albopictus*. Both these mosquitoes are predominant in Southeast Asia and are also responsible for the spread of other arboviral diseases like dengue virus and chikungunya virus. The incidence of dengue has been increasing over the years and this is of concern to public health workers. Simple laboratory tools for the detection of ZIKV is also lacking. In the absence of drugs and vaccine for these arboviral diseases, vector control is the main option for surveillance and control. *Aedes* larval surveys have been

the hallmark of dengue control along with larviciding and fogging when cases are reported. However, we need new paradigms and options for control of these vectors. The current situation in Southeast Asia clearly proves that effective strategies for vector control need to be proactive and not reactive. This will be the way forward to control epidemics of these diseases inclusive of ZIKV until a vaccine becomes available.

Zika virus: An updated review of competent or naturally infected mosquitoes.

PMID: 29145400 | Indexed in PubMed

Zika virus (ZIKV) is an arthropod-borne virus (arbovirus) that recently caused outbreaks in the Americas. Over the past 60 years, this virus has been observed circulating among African, Asian, and Pacific Island populations, but little attention has been paid by the scientific community until the discovery that large-scale urban ZIKV outbreaks were associated with neurological complications such as microcephaly and several other neurological malformations in fetuses and newborns. This paper is a systematic review intended to list all mosquito species studied for ZIKV infection or for their vector competence. We discuss whether studies on ZIKV vectors have brought enough evidence to formally exclude other mosquitoes than *Aedes* species (and particularly *Aedes aegypti*) to be ZIKV vectors. From 1952 to August 15, 2017, ZIKV has been studied in 53 mosquito species, including 6 *Anopheles*, 26 *Aedes*, 11 *Culex*, 2 *Lutzia*, 3 *Coquillettia*, 2 *Mansonia*, 2 *Eretmapodites*, and 1 *Uranotaenia*. Among those, ZIKV was isolated from 16 different *Aedes* species. The only species other than *Aedes* genus for which ZIKV was isolated were *Anopheles coustani*, *Anopheles gambiae*, *Culex perfuscus*, and *Mansonia uniformis*. Vector competence assays were performed on 22 different mosquito species, including 13 *Aedes*, 7 *Culex*, and 2 *Anopheles* species with, as a result, the discovery that *A. aegypti* and *Aedes albopictus* were competent for ZIKV, as well as some other *Aedes* species, and that there was a controversy surrounding *Culex quinquefasciatus* competence. Although *Culex*, *Anopheles*, and most of *Aedes* species were generally observed to be refractory to ZIKV infection, other potential vectors transmitting ZIKV should be explored.

Pathogenesis and Manifestations of Zika Virus-Associated Ocular Diseases.

PMID: 35736984 | Indexed in PubMed

Zika virus (ZIKV) is mosquito-borne flavivirus that caused a significant public health concern in French Polynesia and South America. The two major complications that gained the most media attention during the ZIKV outbreak were Guillain-Barré syndrome (GBS) and microcephaly in newborn infants. The two modes of ZIKV transmission are the vector-borne and non-vector borne modes of transmission. *Aedes aegypti* and *Aedes albopictus* are the most important vectors of ZIKV. ZIKV binds to surface receptors on permissive cells that support infection and replication, such as neural progenitor cells, dendritic cells, dermal fibroblasts, retinal pigment epithelial cells, endothelial cells, macrophages, epidermal keratinocytes, and trophoblasts to cause infection. The innate immune response to ZIKV infection is mediated by interferons and natural killer cells, whereas the adaptive immune response is mediated by CD8+T cells, Th1 cells, and neutralizing antibodies. The non-structural proteins of ZIKV, such as non-structural protein 5, are involved in the evasion of the host's immune defense mechanisms. Ocular manifestations of ZIKV arise from the virus' ability to cross both the blood-brain barrier and blood-retinal barrier, as well as the blood-aqueous barrier. Most notably, this results in the development of GBS, a rare neurological complication in acute ZIKV infection. This can yield ocular symptoms and signs. Additionally,

infants to whom ZIKV is transmitted congenitally develop congenital Zika syndrome (CZS). The ocular manifestations are widely variable, and include nonpurulent conjunctivitis, anterior uveitis, keratitis, trabeculitis, congenital glaucoma, microphthalmia, hypoplastic optic disc, and optic nerve pallor. There are currently no FDA approved therapeutic agents for treating ZIKV infections and, as such, a meticulous ocular examination is an important aspect of the diagnosis. This review utilized several published articles regarding the ocular findings of ZIKV, antiviral immune responses to ZIKV infection, and the pathogenesis of ocular manifestations in individuals with ZIKV infection. This review summarizes the current knowledge on the viral immunology of ZIKV, interactions between ZIKV and the host's immune defense mechanism, pathological mechanisms, as well as anterior and posterior segment findings associated with ZIKV infection.

Zika Virus.

PMID: 27029595 | Indexed in PubMed

Zika virus (ZIKV) is an arthropod-borne virus (arbovirus) in the genus *Flavivirus* and the family *Flaviviridae*. ZIKV was first isolated from a nonhuman primate in 1947 and from mosquitoes in 1948 in Africa, and ZIKV infections in humans were sporadic for half a century before emerging in the Pacific and the Americas. ZIKV is usually transmitted by the bite of infected mosquitoes. The clinical presentation of Zika fever is nonspecific and can be misdiagnosed as other infectious diseases, especially those due to arboviruses such as dengue and chikungunya. ZIKV infection was associated with only mild illness prior to the large French Polynesian outbreak in 2013 and 2014, when severe neurological complications were reported, and the emergence in Brazil of a dramatic increase in severe congenital malformations (microcephaly) suspected to be associated with ZIKV. Laboratory diagnosis of Zika fever relies on virus isolation or detection of ZIKV-specific RNA. Serological diagnosis is complicated by cross-reactivity among members of the *Flavivirus* genus. The adaptation of ZIKV to an urban cycle involving humans and domestic mosquito vectors in tropical areas where dengue is endemic suggests that the incidence of ZIKV infections may be underestimated. There is a high potential for ZIKV emergence in urban centers in the tropics that are infested with competent mosquito vectors such as *Aedes aegypti* and *Aedes albopictus*.

Short Report: Asymptomatic Zika virus infections with low viral loads not likely to establish transmission in New Orleans *Aedes* populations.

PMID: 32469909 | Indexed in PubMed

Aedes aegypti and *Aedes albopictus* are both vectors of Zika virus and both are endemic to the New Orleans Metropolitan area. Fortunately, to date there has been no known autochthonous transmission of Zika virus in New Orleans. No studies of the vector competence of local populations of *Ae. aegypti* and *Ae. albopictus* for Zika virus transmission have been conducted. To determine if New Orleans *Ae. aegypti* and *Ae. albopictus* mosquitoes are competent for Zika virus, mosquitoes were reared to generation F3 from eggs collected in New Orleans during the 2018 mosquito season. Adults were fed an infectious blood meal and kept for 15 days in an environmental chamber. Transmission assays were conducted at 4, 10, and 15 days post exposure and RT-PCR was run on bodies and saliva to detect the presence of Zika virus RNA. We observed remarkably low susceptibility of both *Ae. aegypti* and *Ae. albopictus* from New

Orleans to a Zika strain from Panama after oral challenge. These results suggest a limited risk of Zika virus transmission should it be introduced to the New Orleans area, and may partially explain why no transmission was detected in Louisiana during the 2016 epidemic in the Americas, despite multiple known travel associated introductions to New Orleans. Despite these results these mosquito populations are known to be competent vectors for some other mosquito-borne viruses and control measures should not be relaxed.

Transmission potential of African, Asian and American Zika virus strains by *Aedes aegypti* and *Culex quinquefasciatus* from Guadeloupe (French West Indies).

PMID: 31109248 | Indexed in PubMed

Zika virus (ZIKV) is an arbovirus that has dramatically spread in South America and the Caribbean regions since 2015. The majority of vector incrimination studies available for ZIKV showed that *Aedes aegypti* mosquitoes are important vectors for this virus. However, several reports suggest that *Culex quinquefasciatus* mosquitoes may be implicated in ZIKV transmission in certain urban settings. In the present study, we evaluated the vector competence for ZIKV of *Cx. quinquefasciatus* and *Ae. aegypti* mosquitoes from Guadeloupe using African, American and Asian strains. The results demonstrated that *Cx. quinquefasciatus* is refractory to ZIKV infection whatever the strain tested at 7, 14 or 21 days post-infection (dpi), while ZIKV transmission was recorded in *Ae. aegypti* for all the three strains. The African ZIKV strain was better transmitted by *Ae. aegypti* (~ 50% mean transmission efficiency) and with a shorter incubation period (7 dpi) when compared to the Asian and American strains (<14% transmission efficiency; incubation period of 14-21 dpi). Taken together, these results suggest that only *Ae. aegypti* mosquitoes are involved in urban ZIKV transmission in Guadeloupe and highlight a higher infectiousness of the African ZIKV strain in this mosquito species when compared to the Asian and American ones.

Effects of copper nanoparticles synthesized from the entomopathogen *Metarhizium robertsii* against the dengue vector *Aedes albopictus* (Skuse, 1894).

PMID: 39585856 | Indexed in PubMed

Aedes albopictus, known as the Asian tiger mosquito, is a significant vector for dengue fever, chikungunya, Zika virus, yellow fever. Current control methods rely on chemical insecticides, which face challenges such as resistance, environmental harm, and impact on non-target species *Eudrilus eugeniae* and *Artemia salina*. This study evaluates the toxic effects of biogenic copper nanoparticles (CuNPs) synthesized using *Metarhizium robertsii* intracellular extract obtained from our previous research. The CuNPs were tested against *A. albopictus* and non-target species at 24 and 48 hours post-treatment. Results demonstrated that entomopathogenic fungi-derived CuNPs exhibited potent mosquitocidal activity, resulting in 97.33% mortality in larvae, 93.33% in pupae, and 74.66% in adults at 48 hours post-treatment. The CuNPs derived from *M. robertsii* showed lower LC50 values of 74.873 mg/L in larvae, 76.101 mg/L in pupae, and 136.645 mg/L in adults at 48 hours post-treatment. Additionally, 12 hours post-treatment, catalase (an antioxidant enzyme) activity decreased 1.5-fold in a dose-dependent manner, while glutathione S-transferase (a detoxification enzyme) activity increased 7.8-fold. CuNPs demonstrated lower toxicity to non-target species, with 24% mortality in *A. salina* and 24.44% mortality

in *E. eugeniae* at 24 hours post-treatment. The LC50 values were 634.747 mg/L for *A. salina* and 602.494 mg/L for *E. eugeniae* at 24 hours post-treatment. These findings indicate that entomopathogenic fungi-derived CuNPs are a promising, target-specific candidate for controlling *A. albopictus* at various life stages (larvae, pupae, and adults).

Evidence of Spreading Zika Virus Infection Caused by Males of Different Species.

PMID: 36146853 | Indexed in PubMed

Zika virus (ZIKV) is a positive-sense single-stranded RNA flavivirus and is mainly transmitted by *Aedes* mosquitoes. This arbovirus has had a significant impact on health in recent years by causing malformations, such as microcephaly in babies and Guillain-Barré syndrome in adults. Some evidence indicates that ZIKV can be sexually transmitted and may persist in the male reproductive tract for an extended period in humans. Knockout and vasectomized mice have been used as models to reveal ZIKV infection in the male reproductive tract as a virus source. ZIKV presence in male and female mosquito reproductive tracts and eggs point to venereal and vertical/transovarian transmission, again demonstrating that the reproductive tract can be involved in the spread of ZIKV. Moreover, eggs protected by eggshells have the potential to be a ZIKV reservoir. Given the +/-lack of vaccines and therapies for Zika fever and the underestimated prevalence rate, an understanding of ZIKV infection and its spread from the reproductive tract, which is protected from the immune system and potentially active for virus transmission, is imperative. We must also develop cheaper, more efficient techniques for virological surveillance inside vectors and humans, control vectors with ecofriendly insecticides, and promote condom use to avoid ZIKV contamination during sexual intercourse, as recommended by the World Health Organization.